

SELMA 13 WNW, AL, USA

WMO: 722360

Lat: 32.4567N

Lon: 87.2422W

Elev: 59

StdP: 100.62

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 63858

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-4.1	-11.7	1.4	-2.7	-9.3	1.7	-0.4	5.3	8.2	4.8	7.8	0.8	N/A	0.321

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	35.3	24.5	34.1	24.6	33.0	24.5	26.5	32.1	26.0	31.4	25.6	30.9	1.7	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.3	29.2	24.5	19.7	28.8	24.1	19.1	28.4	82.5	32.2	80.7	31.6	78.8	30.9	28.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
4.3	3.8	3.2	-9.7	36.6	2.2	1.6	-11.3	37.8	-12.6	38.7	-13.8	39.7	-15.4	40.8		
			-10.3	27.5	2.6	0.4	-12.1	27.8	-13.7	28.0	-15.1	28.2	-17.0	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	7.3	9.8	13.7	17.5	21.6	25.8	26.8	26.9	24.3	18.0	11.8	9.3
	DBStd	8.09	5.84	5.61	4.87	3.71	3.18	1.81	1.53	1.92	2.72	4.65	4.78	5.46
	HDD10.0	335	124	68	21	2	0	0	0	0	0	2	34	83
	HDD18.3	1361	342	243	154	58	13	0	0	0	1	65	202	282
	CDD10.0	3176	42	63	137	228	359	473	522	524	428	249	88	63
	CDD18.3	1160	1	5	12	34	113	223	263	266	179	54	6	3
	CDH23.3	10825	4	21	96	328	1041	2155	2491	2525	1582	531	42	11
CDH26.7	4657	0	1	8	49	352	1011	1164	1233	682	156	4	0	
Wind	WSAvg	1.2	1.6	1.7	1.6	1.2	0.9	0.9	0.8	0.9	0.9	1.0	1.2	1.4
Precipitation	PrecAvg	1320	125	127	146	107	95	106	126	105	95	71	96	123
	PrecMax	1718	232	203	296	261	226	240	296	266	260	221	209	270
	PrecMin	699	48	42	29	33	6	29	26	30	16	5	18	62
	PrecStd	234	54	45	76	53	60	62	63	55	55	47	52	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.8	25.7	27.8	29.3	32.9	35.9	35.8	37.8	35.0	32.3	26.9	24.9
		MCWB	18.2	19.7	18.7	21.0	22.8	24.2	25.3	24.3	23.0	21.0	19.2	19.5
	2%	DB	21.6	23.5	26.0	28.1	31.4	34.5	34.5	35.7	33.5	30.1	24.5	22.6
		MCWB	17.2	18.4	18.6	20.1	22.8	24.7	25.5	24.3	23.4	21.0	18.1	18.9
	5%	DB	19.2	21.4	24.2	26.9	30.1	33.2	33.4	34.2	32.2	28.4	22.6	20.7
		MCWB	16.2	17.3	17.7	19.5	22.3	24.6	25.1	24.6	23.4	20.1	17.5	17.6
	10%	DB	17.0	19.3	22.3	25.4	28.9	32.0	32.3	32.6	30.8	26.5	20.7	18.7
		MCWB	14.0	15.8	16.6	18.7	21.8	24.4	24.9	24.7	23.4	19.8	16.2	16.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.7	20.5	20.6	23.0	24.6	26.8	27.1	26.9	26.0	24.5	21.7	21.6
		MCDB	22.2	24.5	24.3	27.4	29.4	32.3	33.0	32.9	30.8	28.1	23.3	23.6
	2%	WB	18.2	19.5	19.7	21.8	23.7	26.2	26.4	26.4	25.2	23.6	20.3	20.0
		MCDB	20.5	23.0	24.1	26.0	28.9	31.8	32.1	31.9	30.3	27.4	22.5	22.1
	5%	WB	16.7	18.4	18.8	20.8	23.1	25.6	26.0	25.9	24.7	22.7	18.9	18.2
		MCDB	18.8	21.2	22.7	24.8	28.3	31.1	31.4	31.2	29.7	26.3	21.4	19.9
	10%	WB	14.7	16.8	17.7	19.9	22.6	25.1	25.5	25.4	24.0	21.2	16.9	16.5
		MCDB	16.5	19.1	21.4	23.7	27.5	30.4	30.8	30.5	28.9	24.6	19.8	18.7
Mean Daily Temperature Range	5% DB	MDBR	12.2	12.6	12.9	13.2	12.5	11.7	10.8	11.2	12.1	14.1	14.5	11.9
		MCDBR	13.1	13.7	13.8	14.1	14.1	13.7	12.5	13.7	13.8	15.2	14.9	12.8
		MCWBR	9.2	8.0	7.2	7.1	6.7	5.4	4.8	4.7	5.4	7.2	9.2	9.7
	5% WB	MCDBR	10.7	10.5	11.3	11.1	11.7	11.8	10.7	10.8	10.8	10.8	11.0	10.8
		MCWBR	8.4	7.2	6.7	6.4	6.0	5.3	4.7	4.5	4.9	5.9	8.3	8.9
Clear-Sky Solar Irradiance	taub	0.323	0.334	0.361	0.399	0.451	0.483	0.545	0.536	0.447	0.374	0.346	0.325	
	taud	2.525	2.498	2.429	2.308	2.214	2.166	2.005	2.025	2.260	2.448	2.468	2.533	
	Ebn at Noon	900	923	918	891	844	813	761	762	821	866	862	877	
	Edh at Noon	85	96	110	130	144	151	176	171	130	100	89	80	
All-Sky Solar Radiation	RadAvg	2.75	3.33	4.47	5.60	6.19	6.27	6.03	5.58	4.96	4.18	3.12	2.37	
	RadStd	0.20	0.42	0.41	0.42	0.42	0.48	0.40	0.33	0.40	0.50	0.31	0.25	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (36 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148	+98

Nomenclature: See separate page